For example: H9 for a hole and d6 for a shaft.

8.8 NEWALL SYSTEM

The Newall system provides a range of clearance, transition and interferance fits for sizes upto 12". It is a hole basic system which stipulates two grades of holes, specified with bilateral tolerances, together with 6 grades of shaft tolerance. It has two grades of bilateral holes (A and B), two grades of press fit (F and D) one transition fit (p) and three clearance fits (Z, Y and X). Hole A is having higher accuracy than B and can be produced by precision reamers.

8.9 BUREAU OF INDIAN STANDARDS

The Bureau of Indian Standards (BIS) is the national standard body of India working under the ministry of consumer affairs, Food and public distribution, government of India. It is established by the Bureau of Indian Standard Act, 1986 which came into effect on 23 December 1986. The Bureau of Indian Standards (BIS), the national standard body of India, resolves to be the leader in all matters concerning standardization, Certification and quality in order to attain this.

This system of limits, fits recommended in IS 919 – 1963 comprises 18 grades of fundamental tolerance or grades of accuracy of manufacturing with designations. IT01, IT0, IT1, …….. IT16. The symbol IT denotes recommended standard tolerance. It also recommended 25 types of fundamental deviations indicated by letters, symbols for holes and shafts. Capital letters of A to Z for holes and small letter of "a to zc" for shafts.

Example

Ø40 H7/gb

It indicates

- Basic size is 40mm
- Hole basic system is adopted
- The tolerance grade of the hole is H7
- The tolerance grade of the shaft is gb.

GLOSSARY

1. Interchangeability – ஒன்றுக்கொன்று பொருந்துதல்
2. Tolerance – ஏற்கப்படும் அளவு வேறுபாடு

ACTIVITIES

1) Visit near industry and look out for sometime in the inspection department.
2) Take and assemble the same make and model of pen parts.

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